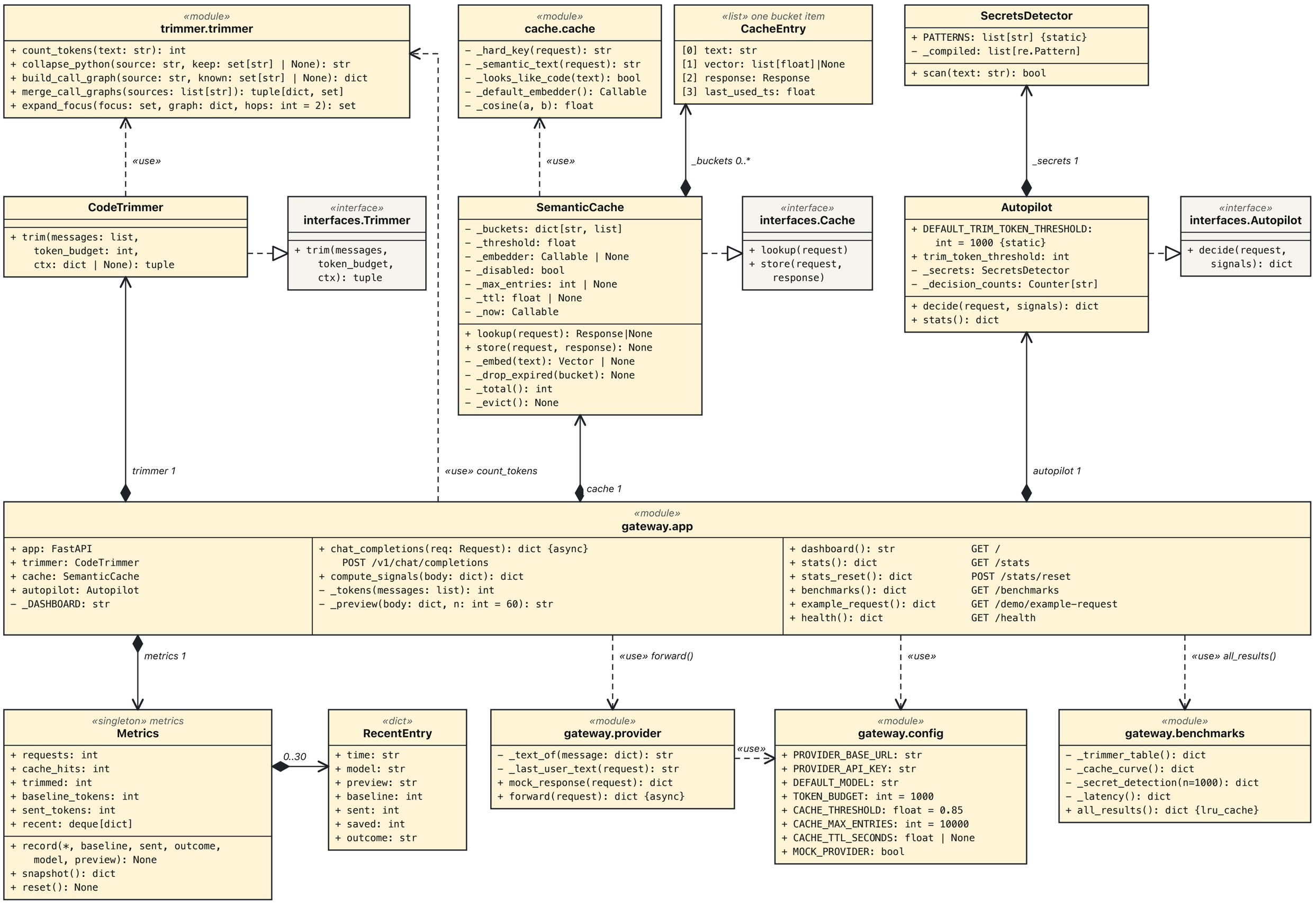
Class Diagram — AI Cost Autopilot

Python backend in code/. Names, signatures and visibility match the source; plain dicts and module-level functions are shown as stereotyped classifiers.



Data shapes passed between classes (plain dicts, not classes)

Request, Response, Message, Stats
Aliases of dict[str, Any] in gateway/interfaces.py (OpenAI chat-completions format).

Signals
num_tokens_est: int
has_code: bool

Plan
use_cache: bool
trim: bool
reason: str

Trim stats
tokens_before: int
tokens_after: int
tokens_saved: int · trimmed: bool

Hard key
SHA-256 of every field except user and message text, plus the message roles.

Notation — realises interface (structural, typing.Protocol) · composition: owner holds the part for its lifetime · dependency «use» · + public - private by convention (leading _)

The gateway uses the concrete classes directly and never imports gateway/interfaces.py; the components match those protocols by shape. The dashboards reach the gateway only over HTTP, so they appear in the sequence and deployment diagrams.